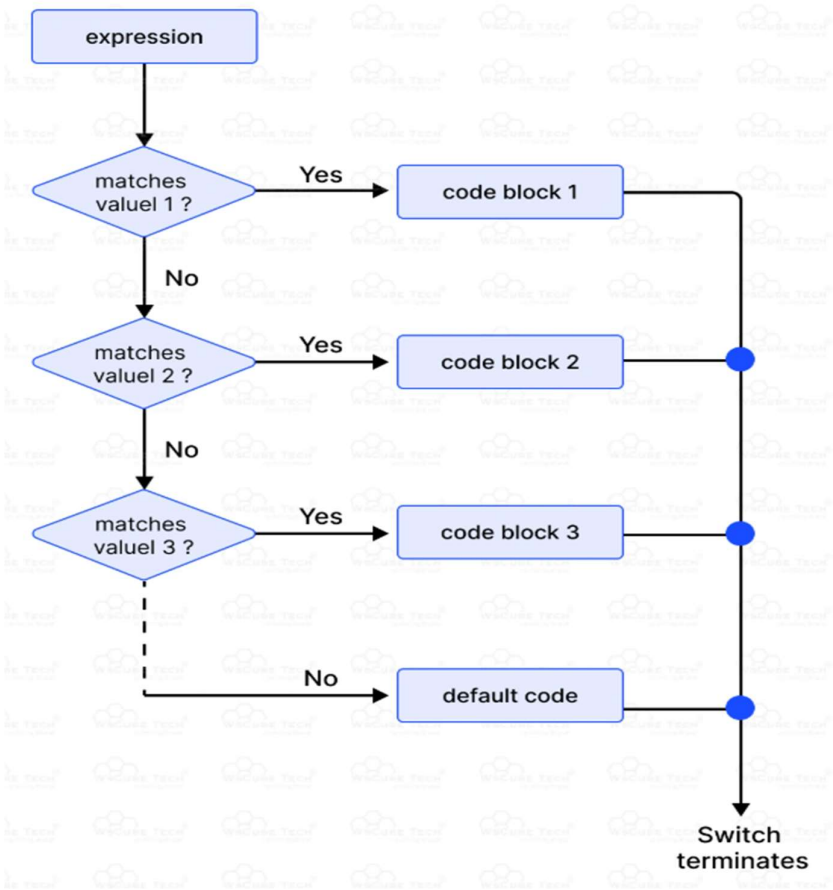
Conditional Statement

Switch Case :-

- Instead of writing many **if..else** statements, you can use the **switch** statement.



Syntax :-

```
switch(2)
{
Case 1:
     Statement1;
     break;
Case 2:
     Statement2;
     break;
Case 3:
     Statement3;
     break;
}
StatementN;
```

Switch Case Implementation :-

Input :-

```java
package basicconcepts;

public class switchstatement {

    public static void main(String[] args) {
        int num=30;

        switch(num)
        {
        case 10:System.out.println("Number is 10");
        break;
        case 20:System.out.println("Number is 20");
        break;
        case 30:System.out.println("Number is 30");
        break;
        default:System.out.println("None");
        }
    }
}
```

Output :-

```
None
```

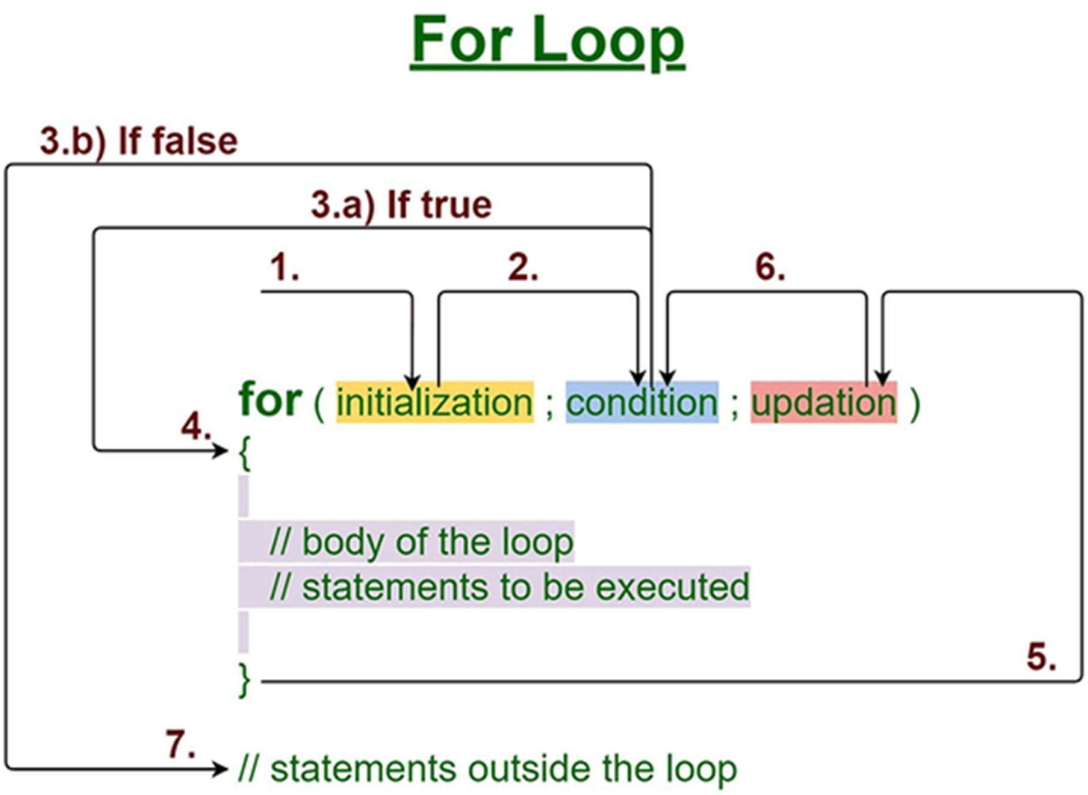
Looping Statement

For loop :-

- The for loop is used when we know the number of iterations.
- The for statement includes the initialization, condition, and increment/decrement in one line.

for loop syntax :-

```
for (initialization; condition; increment/decrement)
{
        // code to be executed
}
```



For loop Implementation :-

Input :-

```java
package basicconcepts;

public class forloopdemo {
    public static void main(String[] args) {

        //Using for loop print name 4 times

        for(int i=1;i<=6;i++)
        {
            System.out.println("Nikhil");

        }
    }

}
```

Output :-

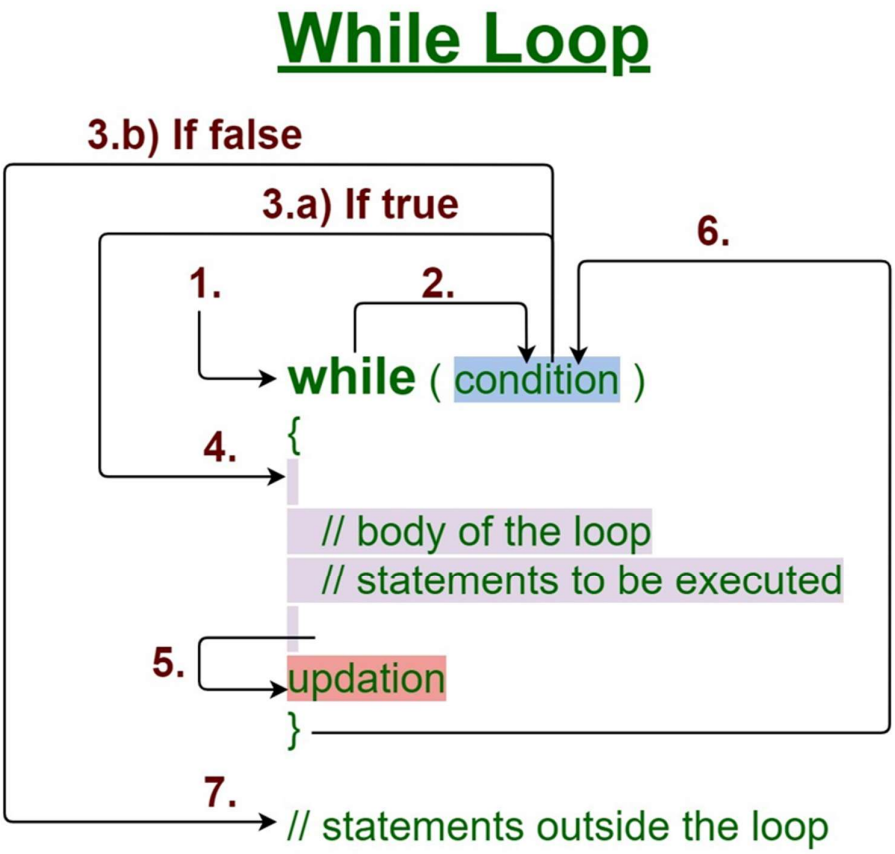
```
Nikhil
Nikhil
Nikhil
Nikhil
Nikhil
Nikhil
```

While loop :-

- A while loop is used when we want to check the condition before executing the loop body.

while loop syntax :-

```
while (condition)
{
        // code to be executed
}
```



While loop Implementation :-

Input :-

```java
package basicconcepts;

public class whiledemo {
    public static void main(String[] args) {

        //print the number for 1 to 10
        int n=1;//initialization

        while(n<=10) //1<=10, 2<=10 3<=10
        {
            System.out.println(n); // 1 2 3 4
            n++;// 2 3 4 ... 10
        }
    }
}

```

Output :-

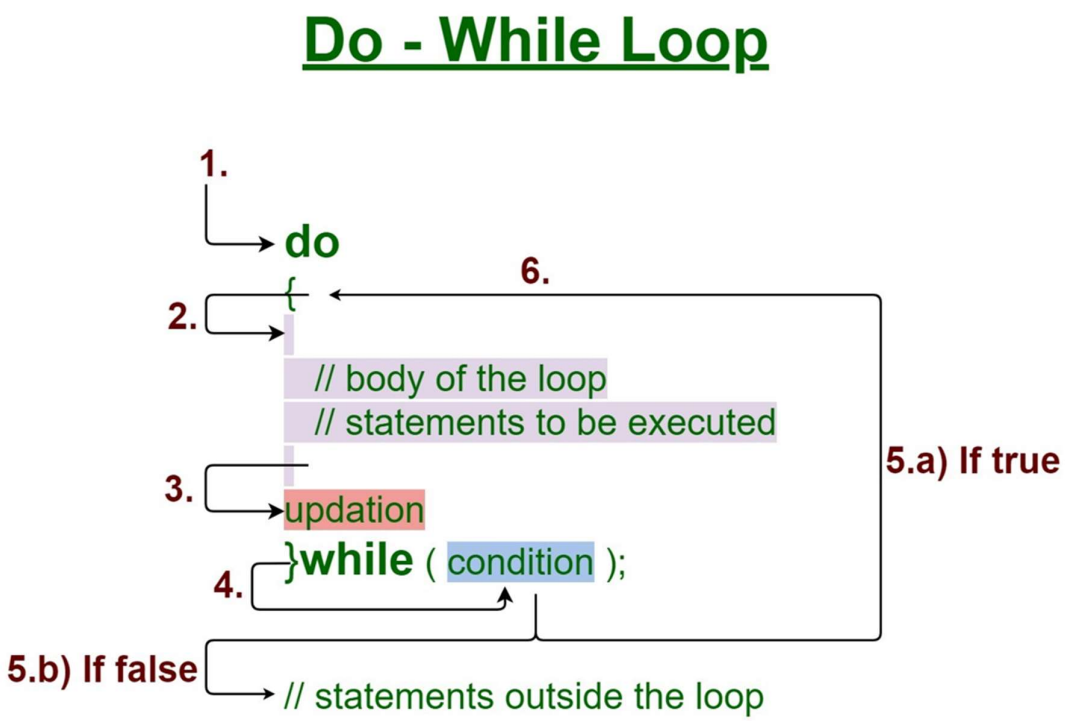
```
1
2
3
4
5
6
7
8
9
10
```

Do-while loop :-

- The do-while loop ensures that the code block executes at least once before checking the condition.

do while loop syntax :-

```
do
{
        // code to be executed
} while (condition);
```



Do-while loop :-

Input :-

```java
package basicconcepts;

public class dowhiledemo {
    public static void main(String[] args) {

        //print 40-45 number
        int num=40;

        do {
            System.out.println(num);
            num++;
        }while(num<=45);
    }
}

```

Output :-

```
40
41
42
43
44
45
```

TASK – 1

Input :-

```java
package assignment;

import java.util.Scanner;

public class task1 {

    //Check Weather the number is odd or even

    public static void main(String[] args) {

        System.out.println("Enter the number: ");
        Scanner sc=new Scanner(System.in);
        int num=sc.nextInt();

        if(num%2==0)
        {
            System.out.println("Given number is EVEN");
        }
        else
        {
            System.out.println("Given number is ODD");
        }
    }
}
```

Output :-

```
Enter the number:
2
Given number is EVEN
```

TASK - 2

Input :-

```java
package assignment;

import java.util.Scanner;

public class task2 {
    public static void main(String[] args) {

        // Check the largest number among two number

        System.out.println("Enter the two number:");

        Scanner sc=new Scanner(System.in);
        int a= sc.nextInt();
        int b= sc.nextInt();

        if(a>b)
        {
            System.out.println(a + "is a largest Number");
        }
        else if (b>a)
        {
            System.out.println(b + " is a largest Number" );
        }
        else
        {
            System.out.print("Equal Numbers");
        }
    }
}
```

Output :-

```
Enter the two number:
20 40
40 is a largest Number
```

TASK – 3

Input :-

```java
package assignment;

import java.util.Scanner;

public class task3 {

    // Check whether user is eligible for voting or not

    public static void main(String[] args) {

        System.out.println("Enter the age:");
        Scanner sc=new Scanner (System.in);
        int age=sc.nextInt();

        if(age>=18)
        {
            System.out.println("You are Eligible for Voting");
        }
        else
        {
            System.out.println("You are not Eligible for Voting");
        }
    }
}

```

Output :-

```
Enter the age:
22
You are Eligible for Voting
```

TASK - 4

Input :-

```java
package assignment;

public class task4 {

    //print 50 to 320 even number

    public static void main(String[]args) {

        for(int i=50; i<320; i++)
        {
            if(i%2==0)
            {
                System.out.println(i);
            }
        }
    }
}
```

Output :-

```
300
302
304
306
308
310
312
314
316
318
```

TASK – 5

Input :-

```java
package assignment;

import java.util.Scanner;

public class task5 {

    // print odd number series using user input

    public static void main(String[] args) {

        Scanner sc=new Scanner (System.in);

        System.out.println("Enter Starting Number:");
        int start = sc.nextInt();

        System.out.println("Enter Ending Number:");
        int end = sc.nextInt();

        for(int i=start; i<=end ;i++)
        {
            if(i%2!=0)
            {
                System.out.println(i);
            }
        }
    }
}
```

Output :-

```
Enter Starting Number:
10
Enter Ending Number:
20
11
13
15
17
19
```